

# Sean Luka Girgis

Observability / APM Platform Engineer | Telemetry, Performance, Python & Reliability

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## Professional Summary

Senior observability, APM, telemetry, and performance engineering professional with 20+ years of enterprise experience across Dynatrace AppMon/Synthetics, CA APM/Introscope, AppDynamics, BMC TrueSight/TSCO, telemetry analysis, dashboards, alerting, production troubleshooting, Python scripting, SQL, cloud data workflows, and system reliability. Strong background supporting large-scale monitoring environments, transforming infrastructure and application telemetry into actionable reliability insights, defining monitoring standards, troubleshooting production issues, and documenting support/runbook practices. Best aligned to observability platform roles requiring telemetry pipelines, metrics/logs/traces concepts, APM platforms, dashboarding, alerting, performance engineering, Python automation, cloud exposure, and production reliability; Prometheus, Grafana, OpenTelemetry, Kubernetes, GPU-cluster observability, and IaC/deployment automation are positioned carefully as adjacent or learning areas rather than overstated production ownership.

## Professional Experience

### Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Supported enterprise observability, capacity, and production reliability workflows across BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, AWS-backed reporting layers, and large-scale infrastructure telemetry feeds.
- Built Python/Pandas/PySpark and SQL workflows that processed telemetry from 6,000+ infrastructure endpoints into validated dashboards, reports, forecasting datasets, and operational insights.
- Analyzed metrics, utilization trends, latency symptoms, missing feeds, abnormal deltas, and reporting discrepancies to help teams identify reliability risk and capacity bottlenecks.
- Developed validation checks for telemetry freshness, row counts, missing data, unmatched records, abnormal metric movement, and output consistency to improve trust in monitoring and reporting data.
- Troubleshoot production issues by tracing source feeds, metric behavior, Python processing, SQL transformations, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Created runbooks, metric definitions, dashboard notes, validation checkpoints, support procedures, and troubleshooting documentation to improve repeatable operational response.
- Partnered with infrastructure, application, analytics, engineering, and business teams to explain telemetry findings, coordinate fixes, and improve production reliability.
- Used AWS S3, Glue, Athena, Redshift, EC2/ECS concepts, Python, SQL, and monitoring data to support cloud-oriented observability and reporting patterns.

### Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, improving visibility into user-facing reliability and performance.
- Correlated real-user monitoring, synthetic transaction data, and load-test behavior to identify bottlenecks, user-impacting symptoms, and improvement opportunities.

- Led the FAST project, mining real-user and synthetic monitoring telemetry to surface optimization recommendations adopted by engineering teams.
- Created dashboards and reporting views that converted raw monitoring signals into actionable performance, reliability, and stakeholder insights.
- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and performance reporting needs.

#### **SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)**

- Served as CA APM/Introscope SME for a large financial-services environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Defined monitoring standards, dashboards, alert rules, SLA thresholds, validation checks, and operational reporting artifacts for application, middleware, infrastructure, and operations teams.
- Provided production troubleshooting guidance using CA APM telemetry, transaction visibility, dashboards, alerts, J2EE/WebLogic monitoring, and performance-analysis patterns.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable observability processes.
- Delivered onboarding support, training material, technical documentation, and knowledge sharing to improve adoption of monitoring and troubleshooting practices.

#### **Performance Test Engineer at AT&T (2010-08 – 2011-07)**

- Analyzed J2EE telecom applications under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and production-readiness testing.
- Documented performance baselines, defects, risks, telemetry findings, and remediation recommendations for engineering and release teams.

#### **Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)**

- Led validation and production-readiness support for a high-throughput shopping engine migration from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.

## **Education**

### **Post-Graduate Diploma in Computer Engineering Technology / Computer Science**

Humber College, Toronto, Canada

### **Bachelor of Science in Civil Engineering**

Zagazig University, Egypt

## Skills

### Flagship Projects

#### Enterprise APM Reporting Automation

Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry across large financial-services monitoring environments.

**Technologies:** CA APM, KornShell/ksh, Perl, SQL, Unix/Linux, Automation

#### FAST Performance Analytics Project

Mined Dynatrace and Gomez Synthetic Monitoring telemetry to identify optimization opportunities, improve transaction visibility, and communicate performance recommendations to engineering teams.

**Technologies:** Dynatrace AppMon, Gomez Synthetic Monitoring, Performance Analytics, Dashboards, APM

#### Production Telemetry Quality and RCA Workflow

Built validation and troubleshooting workflows for telemetry-backed reporting datasets, including freshness checks, missing feed diagnosis, abnormal trend detection, prior-run comparisons, and runbook documentation.

**Technologies:** Python, SQL, Pandas, BMC TrueSight, AppDynamics, RCA, Runbooks

#### ServiceCall AI RAG Demo

Built staged Python proof-of-concepts with FastAPI service layers, Docker-first smoke tests, Pydantic contracts, deterministic endpoints, and testable service behavior.

**Technologies:** Python, FastAPI, Docker, Pydantic, pytest, RAG